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Zinc-Ion Batteries

Lean-Water Gel Electrolyte Enables Zinc Ion Battery at −70 °C

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Abstract: Aqueous zinc ion batteries are attracting growing interest in electrochemical energy storage due to safety, reliability, and affordability. However, water brings drawbacks including parasitic reactions, narrow electrochemical window, and cathode degradation. The freezing nature of water also challenges the zinc ion transport and storage at sub-zero temperatures, especially below −40 °C. Here, we design a water-in-polymer electrolyte to confine water using a weak-solvation monomer-directed polymerization technique initiated by protons. This electrolyte strategy significantly suppresses the water-induced parasitic reactions and widens the electrochemical window to 2.59 V. Besides, a high ionic conductivity of 0.36 mS cm^{-1} is achieved at −70 °C benefiting from unique fast ion transport channel and favorable desolvation process at the interface. Symmetric Zn cells exhibit excellent cycle stability over 10 000 h (437 days) at room temperature (1 mA cm^{-2}@1 mAh cm^{-2}) and 700 h at −40 °C (5.25 mA cm−2@5.25 mAh cm^{-2}). Zn||$Zn_{0.58}V_2O_5$ full cells show impressive performance under high mass loading and low temperatures originating from the suppression of H^+ insertion. The electrolyte strategy in this work will inspire more efforts for water confinement in aqueous batteries.

Introduction

Aqueous zinc ion batteries have been regarded as a promising candidate for next-generation electrochemical energy storage because of their intrinsic high safety, low cost, and environmental friendliness using "water" as electrolyte solvent.[1–3] However, excessive free water in the electrolyte compromises low-temperature performance due to its high freezing point.[4–6] In addition, the highly reactive water tends to induce serious side reactions (hydrogen evolution, corrosion, and passivation) on the zinc metal anode.[7,8] This leads to reduced electrode reversibility and shorter cycle life, greatly hindering its commercialization.

To enhance the stability of zinc anodes, researchers have explored various strategies to reduce water activity, including water-in-salt,[9,10] organic mixing,[11,12] and hydrogel electrolytes.[13,14] Among these strategies, hydrogels have attracted much attention due to their unique advantages. By combining the advantages of both liquid and solid electrolytes, hydrogels can effectively circumvent the high cost and viscosity problems of highly concentrated salts, as well as the lack of stability of organic mixtures. Specifically, hydrogels can effectively regulate the hydrogen bonding network of water through highly integrated chain segments and functional groups, thus inhibiting water activity.[15,16] At the same time, its uniformly cross-linked network structure ensures uniform transport of zinc ions and effectively inhibits the growth of dendrites.[17,18] Therefore, hydrogel electrolytes show significant advantages in inhibiting water activity and ensuring conductivity. However, in order to maintain its high ionic conductivity, the water content in conventional gel electrolytes is usually still higher than 80%. The high-water content in those hydrogels still inevitably causes serious side reactions and the low-temperature operating range only achieves −40 °C (Scheme 1).[19,20] Further reduction of the water content will in turn lead to a sudden drop of ionic conductivity and an increase of interfacial impedance. Currently, although some progress has been made in the design of water-poor gel electrolytes,[14,21,22] they are only suitable for use at room temperature, and their kinetic properties remain limited. This leads to an intractable contradiction between the water content and transport kinetics inside the battery, becoming a critical bottleneck in the development of low-temperature zinc ion batteries. Therefore, there is an urgent need to develop a depleted hydrogel electrolyte system to realize the breakthrough in the key parameter indexes such as low water content and high ion transport.

Here, we propose to construct a novel water-in-polymer, low-temperature-resistant gel electrolyte (Scheme 1) by in situ polymerization of weakly solvated 1,3-dioxolane (DOL) using native protons in the electrolyte. The water content of this electrolyte is reduced to 20% compared to >80% in conventional gels, and the electrochemical window is broadened to 2.59 V due to the drastic reduction of the hydrogen

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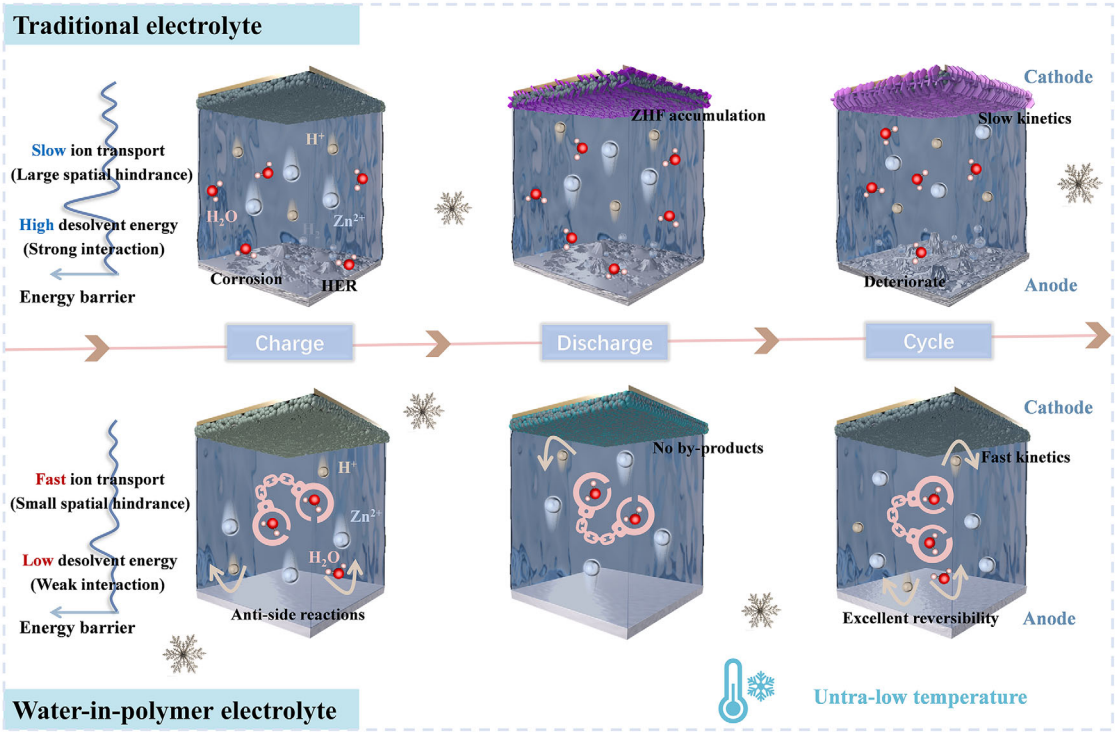

Scheme 1. The schematic diagram of the traditional cross-linked hydrogel and proton-induced gel about the working process of the battery is shown.

bonding network, which significantly enhanced the stability of the electrode. With its unique fast ion transport channel and favorable desolvation process at the interface, the desolvation energy barrier at low temperatures is effectively reduced, demonstrating fast kinetics. The high ionic conductivity of 0.36 mS cm^{-1} can still be maintained under the extreme conditions of −70 °C, realizing a breakthrough in the ultra-low temperature performance of zinc ion batteries. Both symmetric and full cells can be stably cycled for long life under extreme cold conditions.

Results and Discussion

The DOL solvent was gradually added to the 2 m $Zn(OTF)_2$ (ZSF) electrolyte to prepare the water-in-polymer electrolytes, and the corresponding changes were tracked by spectroscopy methods. Fourier-transform infrared (FT-IR) spectroscopy shows that the C—O—C bond strength of DOL gradually weakened with the increase of the solvent ratio (Figure 1a). When the ratio of DOL to H_2O reaches 8:2, C—C polymerization chains appear, indicating that a ring-opening polymerization reaction occurred in DOL and the formation of water-in-polymer electrolyte. Similarly, Raman spectroscopy also suggests this polymerization behavior (Figure S1).[23] At this point, the electrolyte is transformed from liquid to quasi-solid gel electrolyte (GE) (Figure S2), realizing the reconfiguration of the electrolyte.[24] The electrolyte maintains its nonflammable properties, retaining the inherent high safety of zinc ion batteries (Figure S3). Figure 1a also shows that the O—H bending vibration peak of water (1600–1700 cm^{-1}) shifts to higher wavenumbers as the DOL content increases, indicating a disruption of the hydrogen bonding network and restricting the activity of H_2O molecules.[25,26] Molecular dynamics (MD) simulations reveal (Figure 1b) that the hydrogen bonding network of GE is significantly reduced from ∼7800 to 600, which is mainly attributed to the reduction of water content and the restriction of the polymer network. Due to the thermodynamic and kinetic limitations of water molecules, a water-in-polymer environment is formed, effectively suppressing side reactions such as hydrogen and oxygen evolution at the electrode interface.[27] The electrochemical stability window (ESW) is extended from 2.25 to 2.59 V (Figure 1c), which effectively improves the stability of the electrode interface. Then, the formation mechanism of water-in-polymer electrolyte was systematically and thoroughly investigated. In Figure 1d,e, nuclear magnetic resonance (NMR) technology revealed the polymerization process of DOL, and the degree of polymerization of DOL could be quantitatively analyzed by the peak area ratio. As the polymerization time increased, the molecular weight increased, and the conductivity gradually decreased (Figure S4 and Table S1). After 5 days, the electrolyte was basically polymerized, and the system tended to be stable. As shown in Figure S5, comparing the different zinc salt systems, an interesting phenomenon is that the polymerization reaction only occurs in the $Zn(OTF)_2$ system. This difference may originate from the difference in pH. Further experiments showed that introducing acid into pure DOL solvent and high pH zinc salt solution (Figure S5) also triggered polymerization reactions, while raising the pH inhibited the polymerization process. Therefore, we hypothesize

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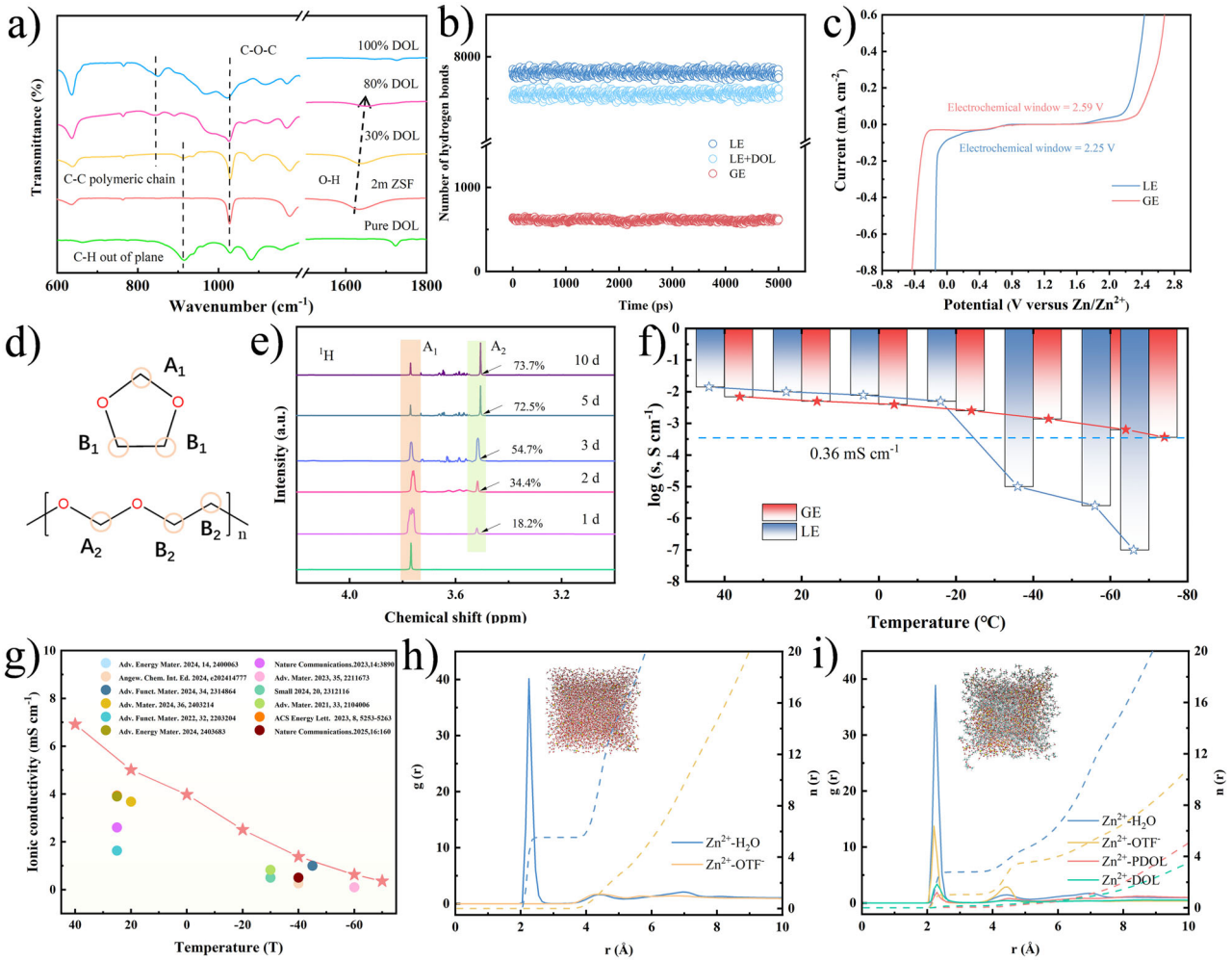

Figure 1. a) The infrared spectra correspond to the introduction of different DOLs. b) MD simulation of hydrogen bond number in the electrolyte (LE, liquid electrolyte; GE, gel electrolyte). c) The ESW test of electrolytes is conducted using the LSV method. d) The chemical structures of DOL and PDOL. e) Corresponding NMR monitoring under different time aggregation conditions(d, day). f) Ionic conductivity tests corresponding to GE at various temperatures. g) The ionic conductivity of GE electrolytes was compared with that of hydrogel electrolytes reported in this paper at different temperatures.[14,17,30–38] h) The radial distribution function of 2 m $Zn(OTF)_2$ electrolyte. i) The radial distribution function of the GE electrolyte.

that the protons in the electrolyte act as initiators, interacting with the lone electron pairs on oxygen in DOL to induce ring opening and initiate the chain polymerization reaction (Figure S6). This process not only effectively eliminates the harmful protons but also avoids the use of additional initiators/crosslinkers. The crosslinker-free short-chain gel system has a low spatial site resistance for ion transport, which enables the ionic conductivity to remain high over a wide temperature range from 40 to −70 °C.[28,29] Even at −70 °C, the ionic conductivity of GE still remains as high as 0.36 mS cm^{-1} (Figure 1f). The GE also remains ice-resistant down to −100 °C (Figure S7). Compared with the previously reported gel system, the GE has a significant advantage in ionic conductivity (Figure 1g). Therefore, this water-in-polymer system not only limits the problems caused by water but also achieves rapid kinetics, laying the foundation for stable low-temperature batteries.

MD calculations were performed based on the actual polymer composition of the electrolyte, and the results showed that PDOL and DOL have weak coordination effects (Figure 1h,i). The weak solvent leads to a decrease in the number of Zn^{2+}-H_2O coordinations from 5.53 to 2.79, which is compensated by additional coordination of the anion. Accordingly, PDOL and DOL do not participate in the primary solvation structure of Zn^{2+} but rather assist in attenuating the strong Zn^{2+}-H_2O interactions (Figure S8). Unlike strongly coordinating solvents that tightly bind to Zn^{2+}, DOL's weak solvation ability facilitates easier desolvation at the electrode interface, reducing the energy barrier for ion transfer. This enhances interfacial ion conductivity, ensuring fast ion transport even at low temperature.[39,40] In summary, the short-chain polymer design of GE not only reduces the content of H^+ and H_2O and locks their activity but also maintains the rapid transport of Zn^{2+}. Importantly, residual DOL can act as a weak solvent, reducing the desolvation energy and ensuring rapid ion transport at the electrode/electrolyte interface.

Zinc metal often leads to uneven distribution of ion flow at the electrode/electrolyte interface due to its rough surface, which in turn triggers the destabilization of the

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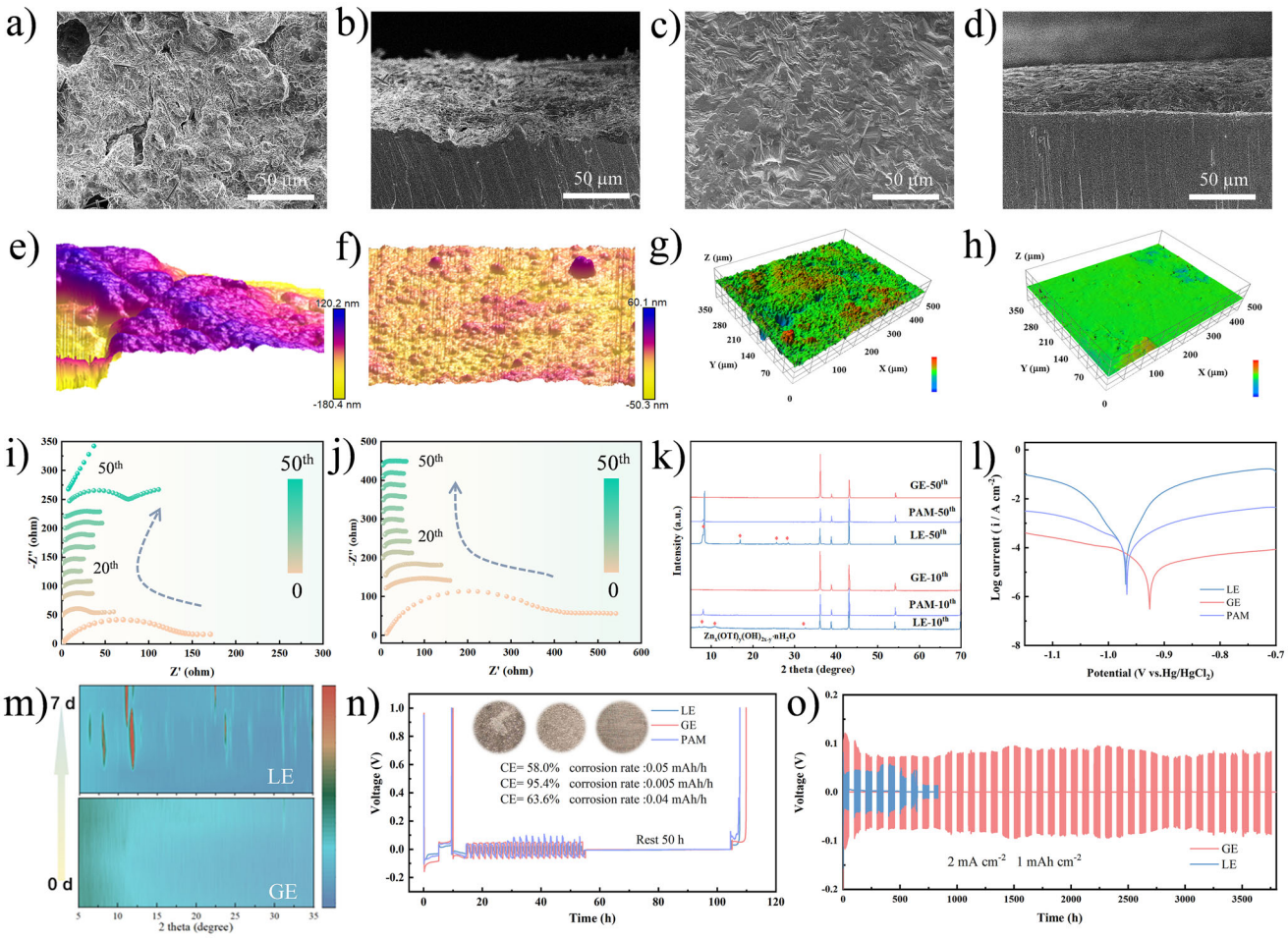

Figure 2. a) Planar and b) cross-sectional SEM images of zinc electrodes after 50 cycles with a symmetric cell of LE. c) Planar and d) cross-sectional SEM photographs of zinc electrodes after 50 cycles with a symmetric cell of GE. AFM images of zinc anode after cycling using e) LE and f) GE at −20 °C. White light interferometer photographs of zinc anode after cycling using g) LE and h) GE, respectively at −20 °C. i) EIS results of batteries at different cycles in LE. j) EIS results of batteries at different cycles in GE. k) XRD patterns of the zinc electrode under different cycle numbers. l) Tafel curve obtained through three-electrode testing. m) XRD patterns of zinc electrodes during the aging process. n) Statistical analysis of the metal corrosion process using the Zn-Cu cell. o) Periodic shelving-cycling performance of symmetric cells.

zinc anode (Figure S9).[41] Scanning electron microscope (SEM) examination of the zinc surface after 50 cycles (Figure 2a,b) reveals the presence of a large number of mossy protrusions and pits, which are attributed to the severe dendrite growth and side reactions in the liquid electrolyte.[42] Similarly, electrodes using polyscrylamide (PAM) gel electrolytes exhibited a rough morphology after cycling (Figure S10). In contrast, the zinc anode employing GE shows a dense morphology after cycling (Figure 2c,d), which is attributed to the promotion of homogeneous transport of Zn^{2+} by the GE chain segments. Combined with the fact that during Zn^{2+} deposition, the increase in nucleation overpotential by GE facilitates grain refinement (Figure S11),[43] the dense zinc deposition can also be obtained. Structural examination of the zinc anodes after cycling at −20 °C by atomic force microscope (AFM) (Figure 2e,f) shows that zinc with GE is smoother and flatter than using LE. The zinc anodes with LE show a rough and disordered surface at −20 °C, confirming its poor low-temperature adaptability. Further lowering the temperature to −40 °C (Figure S12), zinc with GE still achieves a smooth morphology. For LE, the limited solidification point of the electrolyte results in inoperability. Thermal infrared imaging observation of the electrodes after low-temperature cycling reveals that GE exhibits a more uniform temperature distribution compared to LE (Figure S13), which facilitates the realization of uniform heat dissipation and temperature distribution.[43,44] This characteristic also enhances the low-temperature performance of GE. White light interferometer recordings of the three-dimensional structure of the electrodes after cycling yield consistent results (Figure 2g,h). The above results demonstrate that GE significantly improves the stability of the zinc anode at both ambient and low temperatures. Electrochemical impedance spectroscopy (EIS) is further used to monitor the interface changes at different cycles.[45] For LE (Figure 2i), the impedance at high frequency decreases when increasing cycle numbers before 20 cycles, which is ascribed to the increase in the active surface area due to the growth of dendrites. However, the impedance increases after 20 cycles, due to the accumulated by-products blocking charge transfer

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at the zinc anode/electrolyte interface, and X-ray diffraction (XRD) results of cycled zinc anode confirm the formation and accumulation of by-product $Zn_x(OTF)_y(OH)_{2x-y}\cdot nH_2O$ (ZHF) with increasing cycles (Figure 2k), which also leads to a short cycle life in LE. Although PAM electrolytes exhibit low water content compared to LE, they also accumulate ZHF. In contrast, the zinc anode using GE shows stable impedance at high frequency throughout the cycling tests (Figure 2j), thus ensuring long-term stable operation.

The side reactions pose a major challenge to electrode stability.[46] The free water molecules in the electrolyte not only exacerbate electrode corrosion but also promote the rapid transfer of H^+, accelerating the electrode deterioration process.[47,48] GE showed higher corrosion potential and lower corrosion current on the Tafel curve by reducing the water content from 100% to 20% and consuming some H^+ (Figure 2l), which proves that water is effectively trapped by the polymer, leading to its decomposition reaction being effectively suppressed at the thermodynamic and kinetic levels.[49] Immersion of the electrodes in electrolyte systems and XRD/SEM monitor during aging reveal (Figures 2m and S14) that the aging in LE led to a gradual enhancement of the ZHF by-product peaks, accompanied by significant flake byproduct formation. GE shows almost no by-product observation in XRD/SEM, which also indicates its ability to inhibit corrosion from water. Quantitative analysis shows (Figure 2n) that the corrosion rate in LE is 0.05 mAh h^{-1}, resulting in a Coulombic efficiency (CE) of only 58% and a significant active zinc depletion due to by-product formation. PAM electrolyte also exhibits low CE, with a corrosion rate of 0.04 mAh h^{-1}. In contrast, the GE corrosion rate is as low as 0.005 mAh h^{-1}, which is 100 times slower and results in a CE of 95.4%. By simulating actual battery operation through a cyclic cycle-shelf cycle (Figure 2o), GE dramatically extends the battery cycle life from 650 to 3800 h, strongly demonstrating its excellent side-reaction inhibition and stability improvement (Figure S15).

In order to deeply evaluate the effect of GE on the reversibility of zinc anodes, we performed long-cycle tests of symmetric cells.[50] The test results show that the stability of the cell is significantly improved with the decrease of water content in the electrolyte and the increase of the DOL ratio (Figure S16). This change suggests that the suppression of side reactions has a positive effect on enhancing the reversibility of the zinc anodes. However, it is worth noting that the anhydrous system, while improving the stability, also brings higher polarization and limited cycle life constraints. This enlightens us on the necessity of reducing water content while enhancing kinetics. At a DOL ratio of 80%, the cell with GE can cycle stably for more than 10 000 h (437 days) under 1 mA cm^{-2}–1 mAh cm^{-2} cycling conditions (Figure 3a), while cells using LE and PAM have lifespans of 50 and 100 h, respectively. This comparison highlights the 200-fold and 100-fold improvement of cycle life in symmetric Zn cells using GE. In addition, improving zinc anode utilization is critical to increasing battery energy density and reflecting the true reversibility of the electrodes.[51,52] With a zinc utilization of 70% and a zinc foil thickness of 10 microns (Figure 3b), the cell using GE is able to cycle stably for 3500 h. In contrast, for cells with LE, the depletion of active zinc by by-products during cycling (Figure S17) results in a dramatic increase in polarization due to zinc insufficiency and eventual failure of the cell after 25 h. PAM electrolyte slows down side reactions to a certain extent, extending battery life to 65 h. When testing the rate performance of the battery, we find that the GE exhibits a stable cycling curve at different current densities (Figure S18) from 1 to 20 mA cm^{-2}. In contrast, the LE cell becomes unstable in deposition/stripping behavior when the current increases to 10 mA cm^{-2}. Zn-Cu half-cells were assembled with different electrolytes for further investigation. Under 2 mA cm^{-2}–1 mAh cm^{-2} conditions (Figures 3c and S19), the cell with GE exhibits a stable voltage profile with an average Coulombic efficiency of 99.78% after 4500 cycles. In contrast, the cells with LE and PAM have shorter lifespans and lower CE of 41.49% and 72.39%, respectively. Observation of the copper electrodes after cycling reveals (Figure S20) that the surface of the cell using LE is attached with a large number of by-products caused by uneven zinc deposition. For cells using GE, the deposited zinc on the copper surface is completely stripped off, thus contributing to a higher CE. This can be attributed to the higher zinc nucleation potential in GE, resulting in denser zinc deposition (Figure S21) and effectively suppressing the loss of active zinc. Even when the current density is further increased to 5 mA cm^{-2}, the cell with GE still exhibits a high CE and stable voltage profile (Figure S22), demonstrating its effective protective effect.

The low-temperature performance was first evaluated by symmetric cells.[53] At –25 °C and 10 mA cm^{-2}–5 mAh cm^{-2} conditions (Figure S23), the cell with GE shows excellent stability and can be stably cycled for 5000 h. At −40 °C, the cell with GE can be cycled for 700 h even with a zinc utilization rate of up to 80% (Figure 3d), while LE and PAM do not work due to solidification. The variable temperature impedance test shows (Figures 3e and S24) that the activation energy of GE is 22.45 kJ mol^{-1}, which is much lower than that of LE's 37.41 kJ mol^{-1}. The decreased activation energy suggests a faster desolvation process for GE.[54,55] At low temperatures, desolvation, as a key step of the reaction, is critical to the battery performance.[56] The short-chain structure of GE reduces the spatial site resistance and promotes rapid ion transport, while its weak solvation environment ensures a fast desolvation process, thus guaranteeing a good low-temperature performance. Under extreme −70 °C conditions, GE still maintains a stable voltage operating profile (Figure 3f) with a lifetime of 1000 h, demonstrating excellent low-temperature performance. Compared to conventional cross-linked polymerized gel systems (Figure 3g and Table S2), GE maintains the thermodynamic stability of zinc metal by confining water molecules, while the low spatial potential resistance ensures a high ionic conductivity, and the weak solvation environment further enhances its low-temperature performance. In terms of cycle life and discharge depth, GE demonstrates significant advantages over previous reports in all aspects.

To demonstrate the potential of GE in full cells, ion-inserted vanadium oxide was chosen as the cathode material for in-depth investigation (Figures 4a and S25).[57] At a

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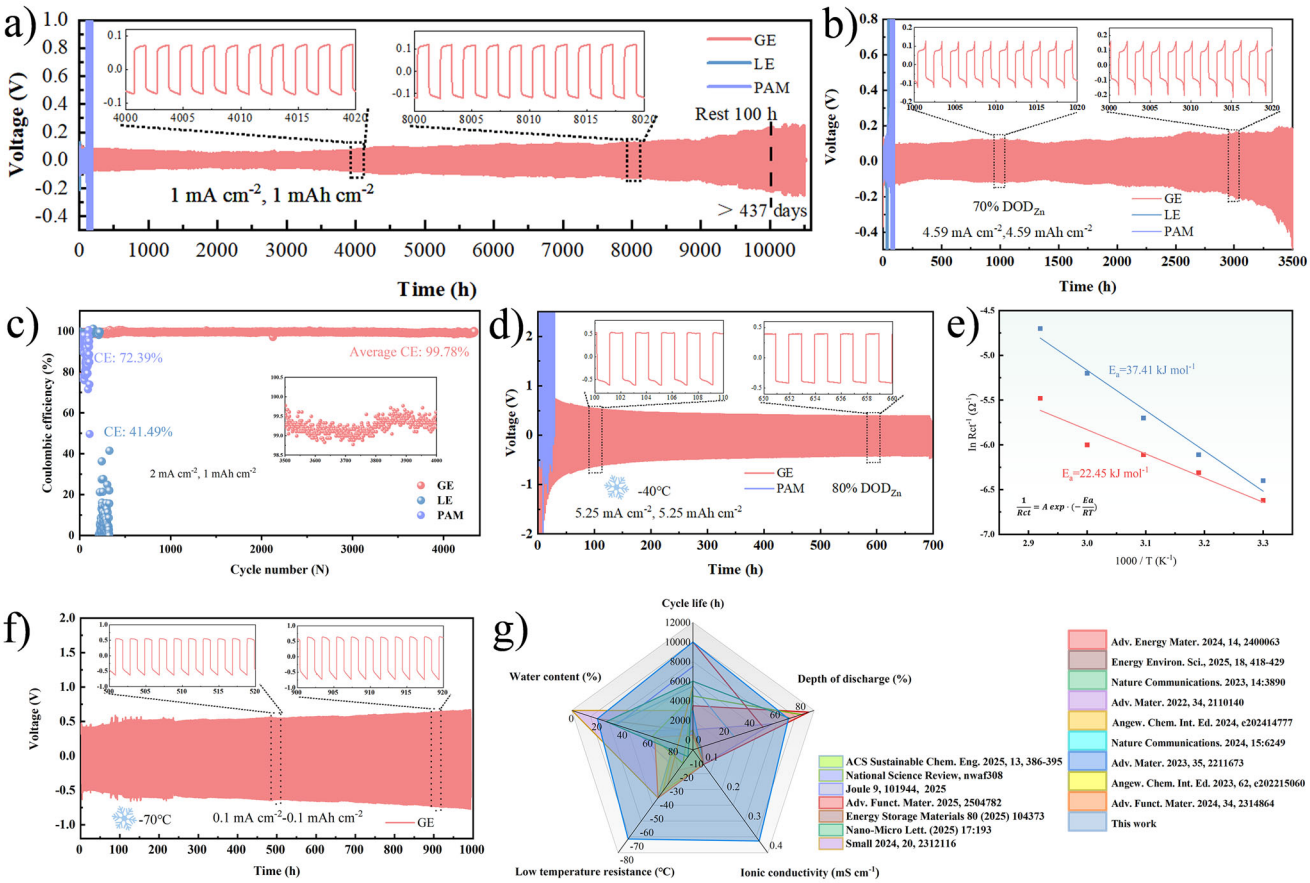

Figure 3. a) Cycling performance of symmetric cells at 1 mA cm^{-2}–1 mAh cm^{-2}. b) Performance testing of symmetrical cells at 70% zinc utilization. c) CE performance of Zn-Cu half-cells at 2 mA cm^{-2}–1 mAh cm^{-2}. d) Cycling performance of symmetrical cells using GE electrolyte at 80% zinc utilization at −40 °C. e) Activation energy fitting curve. f) Symmetrical cells cycling performance at −70 °C using GE electrolyte. g) Radar charts for comparison with other literature on critical parameters.

current density of 1 A g^{-1}, GE demonstrates impressive cycling stability when the cathode material is loaded up to 7.9 mg cm^{-2} (Figure 4b,c). After 600 charge/discharge cycles, its capacity retention remains at 81.5%, demonstrating a stable capacity-voltage curve. In contrast, although the LE exhibits a slightly higher specific capacity in the initial stage, its capacity rapidly declines after only 100 cycles (Figure S26), with a capacity retention of only 11.5% after 600 cycles. Similarly, the use of PAM electrolytes exhibits rapid capacity decay. The morphology observation after 100 cycles (Figure 4d,e) reveals that the zinc anode using LE shows significant roughening compared to GE, which directly reflects the detrimental effect of high active H_2O/H^+ on electrode stability, leading to irreversible zinc loss and thus affecting the overall stability of the cell.[58] By further increasing the current density to 5 A g^{-1} or even 20 A g^{-1}, GE achieved a cycle life of over 9000 cycles. In contrast, the use of LE and PAM electrolytes led to severe side reactions at the negative electrode and material dissolution problems caused by high water content at the positive electrode, resulting in rapid capacity degradation (Figure S27). This effect is especially significant under low-rate conditions, which has become a key factor restricting the development of zinc ion batteries.[59] Under the condition of 0.2 A g^{-1}, GE also shows remarkable cycling stability with almost negligible capacity decay after 200 cycles (Figure 4f). LE, on the other hand, exacerbates the side reaction due to the prolonged working time and promoted the growth of dendrites, leading to the short-circuit failure of the full cell after only 35 cycles (Figure S28). Although the battery using PAM electrolyte will not short-circuit within 200 cycles, it exhibits faster capacity degradation. In addition, the GE's cycling stability at low temperatures is equally impressive. At −20 °C and an electrode loading of 4 mg cm^{-2} (Figure 4g), the full cell with GE is able to cycle stably for more than 500 cycles (at 0.5 A g^{-1}). In contrast, the performance of LE and PAM at low temperatures is significantly poorer, exhibiting low and fluctuating specific capacities. Even at the more severe −40 °C, GE demonstrates excellent rate performance (Figure 4h). As the current density decreases, the capacity climbs. When the current density is reduced to 0.05 A g^{-1} and the load is as high as 7.6 mg cm^{-2}, the cell still releases a capacity of about 160 mAh g^{-1}, indicating its good kinetic performance at low temperature. In the long cycle test at −40 and −70 °C, GE also maintains good cycle stability (Figure 4i), and the performance remains stable after 100 cycles. This fully demonstrates the excellent low-temperature resistance effect of GE, which not only ensures the stability of the zinc electrode but

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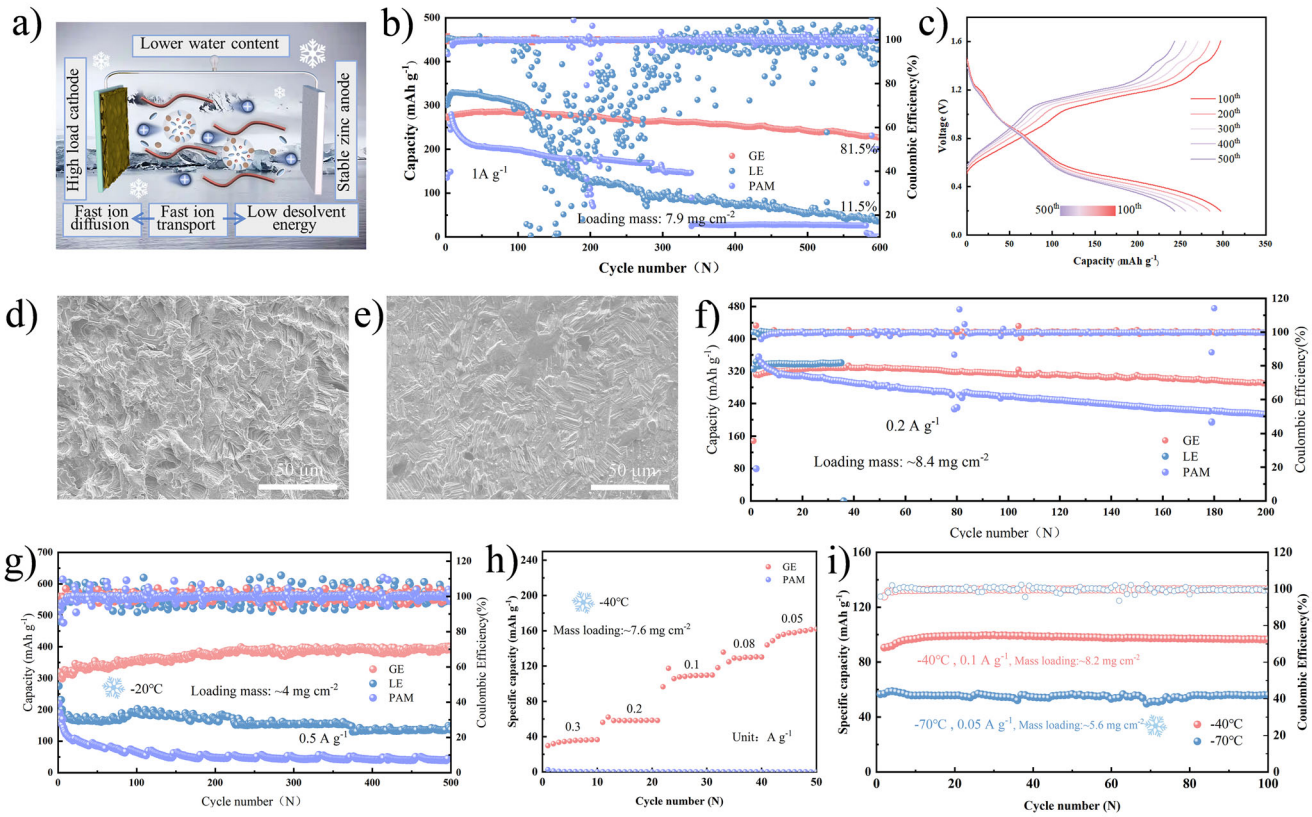

Figure 4. a) Schematic diagram of full cell operation. b) Cycling performance of Zn-$Zn_{0.58}V_2O_5{\cdot}H_2O$ (ZVO) full cells at 1 A g^{-1} and c) the corresponding voltage-capacity curves at different cycles using GE. SEM images of zinc anodes after 100 cycles in d) LE and e) GE electrolytes. f) Long cycle performance of the full cells at 0.2 A g^{-1}. Low-temperature testing of full cells: g) long-cycle stability at −20 °C; h) rate performance at −40 °C; i) long-cycle performance at −40 °C (0.1 A g^{-1}) and −70 °C (0.05 A g^{-1}).

also realizes the excellent performance of the zinc ion full batteries.

To gain a deeper understanding of the ion storage mechanism in the cathode material, we monitor the material changes at different discharge/charge states (Figure 5a). As shown in Figure 5b, the 26° peak of ZVO is shifted to the right during the discharge process, revealing the lattice contraction due to electrostatic interactions with the material skeleton during Zn^{2+} embedding.[60] Meanwhile, the appearance of the ZHF peak is observed at a low-angle region, which is attributed to the accumulation of interfacial OH^- during H^+ embedding and forming ZHF. The intensity of this peak increases significantly when discharged to 0.2 V. Entering the charging stage, the 26° peak of ZVO is shifted to the left and returns to its original shape, indicating the detachment of Zn^{2+}. Accompanying this process, the ZHF peak disappears, reflecting the H^+ detachment, which constitutes a typical Zn^{2+}/H^+ co-embedding mechanism (Figure S29).[61] When using PAM gel electrolytes, consistent ion storage patterns are observed (Figure S30). In contrast, when using GE, ZVO exhibits only peak shifts during discharge/charge without the appearance of new peaks (Figure 5c), suggesting that GE inhibits the embedding/de-embedding of H^+, involving only the de-embedded storage of Zn^{2+} (Figure S29). This is attributed to the fact that GE polymerization consumes the H^+ concentration, while its water-confined environment reduces the number of hydrogen bonds and blocks the transfer of protons through the Grotthus mechanism[62] thus effectively inhibiting the storage of H^+ in the material. It is well-known that the cathode material tends to be covered by ZHF during the H^+ embedding process (Figures 5d and S31), and ZHF significantly hinders further embedding of subsequent ions due to its low ionic conductivity, accompanied by increased interfacial impedance (Figures S32 and S33).[63] This phenomenon is particularly significant when the material loading increases, resulting in the low utilization of active material. Therefore, the H^+/Zn^{2+} storage mode makes it difficult to meet the demand for high loads in real batteries. In order to deeply understand the difference between the two storage modes, we performed finite element simulations. The results show (Figure 5e) that at 20% depth of discharge (DOD), the two materials exhibit similar Zn^{2+} concentrations. However, as the discharge depth increases, the difference between the two is gradually revealed. When reaching 80% DOD, the material with GE exhibits a higher ion concentration inside, indicating unimpeded ion diffusion and excellent kinetics. In contrast, the material using LE is limited by the coverage of ZHF, and the coverage problem becomes more serious with the deepening of H^+ embedding. This results in a large number of Zn^{2+} vacancies within the material at 80% DOD and poor utilization of the active material when using LE. By performing the constant current

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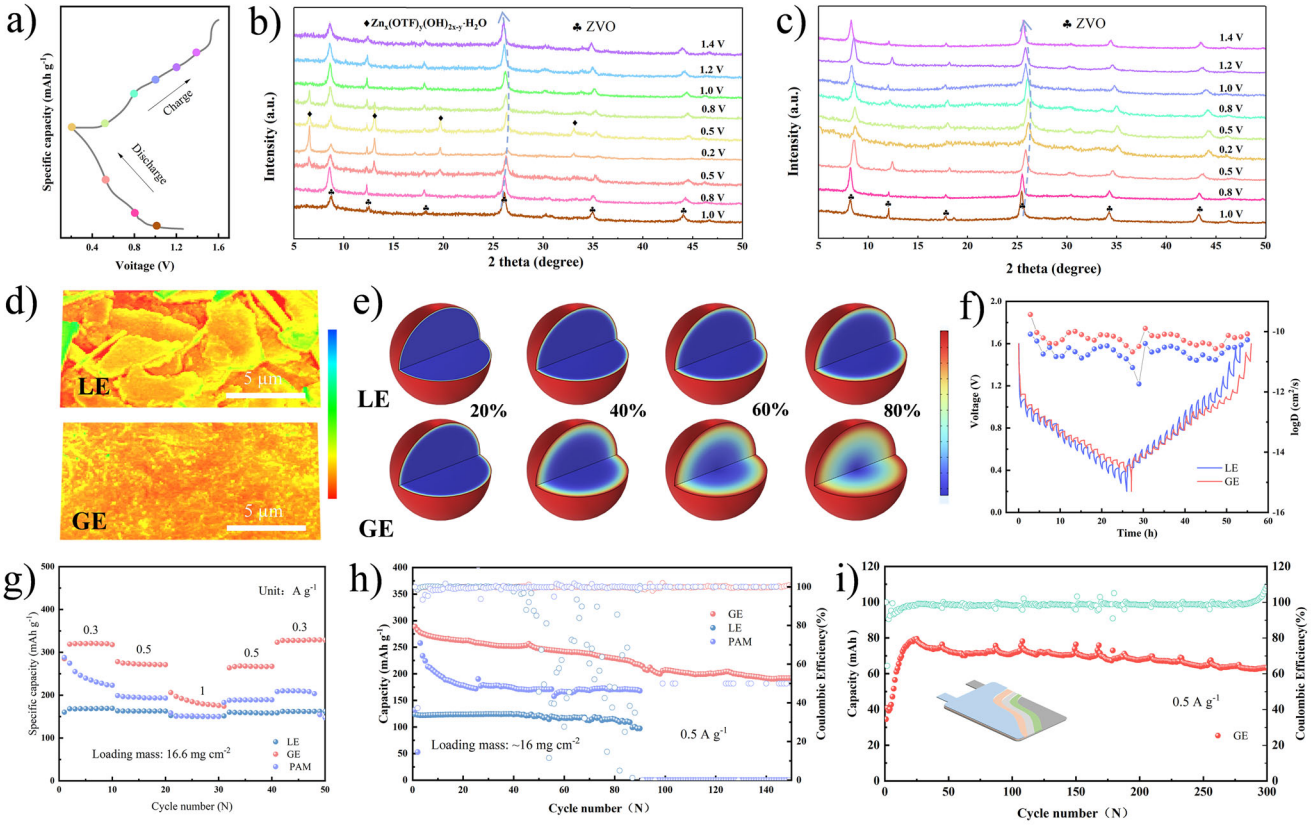

Figure 5. a) Charge–discharge operating curves for full batteries. XRD results of cathode materials corresponding to b) LE and c) GE electrolytes in different charging and discharging states. d) Morphology of cathode electrode material after cycling (the color difference indicates different roughness). e) Finite element simulation of the concentration distribution of zinc ions in different DOD states. f) GITT curves and the corresponding diffusion coefficients of zinc ions. g) The rate performance of the full battery under high loading of the cathode. h) Cycle performance of full battery at 0.5 A g^{-1}. i) Cycle performance of pouch cells at 0.5 A g^{-1} using GE.

intermittent titration technique (GITT) test (Figure 5f), we find that the material in GE has a faster ion diffusion rate compared to LE. This further suggests the accumulation of ZHF, which hinders the ion diffusion in the cathode.[26] When the accumulated ZHF on the cathode surface is removed by chemical treatment, the specific capacity of the cathode shows an increase, further demonstrating the hindering effect of ZHF on the cathode capacity (Figure S34).

Based on the above analysis, we further evaluated the full cell performance under higher mass loading. Under the loading condition of 16.6 mg cm^{-2}, we compare the rate performance of materials in GE, PAM, and LE (Figure 5g). The results show that the cathode using GE delivers a higher specific capacity compared to LE and PAM. Cathode using GE releases 200 mAh g^{-1} even at a current density of 1 A g^{-1}. In addition, Figure 5h shows the cycling performance at a current density of 0.5 A g^{-1}. Cathode in GE exhibits higher capacity and longer cycle life, maintaining a reversible specific capacity of 192 mAh g^{-1} after 150 cycles. On the other hand, the cathode in LE shows a lower capacity, which is accompanied by large fluctuations in the Coulombic efficiency when cycling more than 40 cycles, as well as showing overcharging of the cell due to side reactions leading to cell failure (Figure S35). PAM electrolytes offer intermediate capacity performance, but battery failure occurs earlier at 90 cycles. Finally, we display a soft pack battery using GE (Figures 5i and S36), under conditions of a load of 25 mg cm^{-2} and a current density of 0.5 A g^{-1}, the battery remained stable for over 300 cycles, which fully demonstrates the practical potential of GE.

Conclusion

In summary, we successfully developed a water-in-polymer, low-temperature-resistant, and highly stable gel electrolyte by a weakly solvated DOL-directed polymerization strategy. The water molecules in the electrolyte are confined by polymers and a wide electrochemical window of 2.59 V is achieved, ensuring the electrode stability and side reactions suppression. The electrolyte achieves fast transport and desolvation of zinc ions by virtue of its unique fast ion transport channel and weak solvation environment at the interface, maintaining a high ionic conductivity of 0.36 mS cm^{-1} at −70 °C. The symmetric cells with GE show a cycle life of more than 10 000 h at 25 °C (1 mA cm^{-2}–1 mAh cm^{-2} conditions) and can be cycled for 3500 h even at a discharge depth of 70%. It can also be stably cycled for more than 1000 h at the extreme temperature of −70 °C. In addition, the water-in-polymer electrolyte revolutionizes the cathode ion

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storage mechanism by suppressing the H^+ co-intercalation layer in full cells. This mechanism avoids the generation of insulating ZHF, thus realizing the efficient utilization of cathode materials under high mass loading. The specific capacity of 192 mAh g^{-1} is maintained after 150 cycles at a current density of 0.5 A g^{-1} under a high load of 16 mg cm^{-2}. This work provides new insights for the design of electrolytes to adapt low temperatures, which is important for promoting the application of batteries under extremely cold conditions.

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Conflict of Interests

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords: Low temperature • Water confinement • Zinc anode • Zinc ion batteries

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